Harsh Nandan Shukla

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Education

Indian Institute of Technology Jodhpur — Jodhpur, Rajasthan

Bachelor of Technology, Major in Materials & Metallurgical Engineering — *CGPA: 7.53* *Aug 2023 – Present*

Minor in Artificial Intelligence

Experience

Summer Research Intern – Indian Institute of Information Technology Allahabad (IIIT Allahabad)
Exploring GAN based Defense Strategy for Adversarial Images using Vision Transformer *June–July 2025*
Mentor: Prof. Shivram Dubey, Computer Vision and Biometrics Lab, IIIT Allahabad

- Trained a **Vision Transformer** on **CIFAR-100** (**60,000 images**, 100 classes; **74.17%** clean accuracy) and evaluated robustness under **PGD** adversarial attacks, observing accuracy collapse to **6.66%**.
- Designed a **U-Net-based GAN** reconstruction pipeline with a custom multi-objective loss (adversarial + MSE + cross-entropy), recovering perturbed-image accuracy to ∼**27% (4× improvement)** across **9 experimental phases**.

Data Science Intern – Pull Logic *Sept 2025 – Oct 2025*

- Collected, cleaned, and standardized large-scale **geospatial** and **market datasets** using **Python** and **SQL** (SSMS, Microsoft SQL Server), building extraction, transformation, and validation workflows.
- Performed **statistical analysis** and **hypothesis testing** on inventory and sales trends across provinces and counties; automated querying and visualization in Python to produce client-specific analytical reports for cross-functional teams.

Research Intern (Ganpat University) – Sentiment & Sarcasm Detection ML/DL *April–May 2025*

- Implemented and benchmarked **5 ML models** (Logistic Regression, SVM, NB, Random Forest, XGBoost) and **3 deep learning models** (GRU, BiLSTM, Stacked LSTM) for sarcasm detection on a **1.3M-comment Reddit dataset**.
- Applied comprehensive preprocessing (stemming, stopword/emoji removal, spell correction) and **TF-IDF** vectorization across **800K+ training samples**. Logistic Regression achieved the best performance (**67.3% accuracy, F1 67.25%**).

Projects

LLM Guardrails Gateway API Docs Live Demo *June 2026*

- Built and deployed a production **LLM Guardrails Gateway** wrapping **GPT-4o mini** with a **6-stage fail-fast pipeline** (prompt injection detection, input toxicity check, PII redaction, LLM call, output toxicity check, output PII redaction); containerized with **Docker Compose** orchestrating **FastAPI**, **Streamlit**, **PostgreSQL**, and **Redis**, deployed on **Oracle Cloud Infrastructure (OCI)** behind an **Nginx HTTPS** reverse proxy.
- Benchmarked four guardrail components — **Presidio** PII detection: Precision 0.96 / **Recall** 0.84 vs regex baseline Precision 0.62 / Recall 0.58 via threshold tuning; **Redis** caching: 92% hit rate, latency from **60ms to 47ms** under **Locust** load testing; prompt injection detector (heuristic + **GPT-4o mini** judge): F1 0.97, FPR 0.0 on 107 adversarial samples; **Toxic-BERT** toxicity classifier: **Recall** 0.74, Precision 1.0, F1 0.85 at 128ms.

AgentScope – LLM/Agent Evaluation Platform API Docs Live Demo *Aug. 2026*

- Engineered an **LLM/agent evaluation platform** using **LangGraph** and **LangSmith**, combining loop, budget, schema, and **LLM-as-a-Judge** signals into a unified run-level scoring pipeline with automated trace ingestion and **PostgreSQL** persistence.
- Built and deployed a **FastAPI** + **Streamlit** evaluation service as a multi-container **Docker Compose** stack on **Oracle Cloud**, with IP-based rate limiting and an **Nginx/HTTPS** reverse proxy as the single public entry point, enabling API-driven agent evaluation, run history, and persistent scoring.

Skills

- **Programming**: Python, C/C++, SQL
- **AI/ML**: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Computer Vision, Vision Transformers (ViT), GANs, NLP
- **LLM & GenAI**: LangChain, LangGraph, RAG Pipelines, Prompt Engineering, OpenAI API (GPT-4o), Hugging Face Transformers, Qdrant
- **AI Dev Tools** : Claude Code
- **Backend & APIs**: FastAPI, REST APIs
- **Databases & Caching**: PostgreSQL, Redis
- **Cloud & DevOps**: Docker, Docker Compose, Oracle Cloud (OCI), AWS EC2, Nginx, Git, GitHub, Linux

Relevant Courses

- Artificial Intelligence, Deep Learning, Data Structures & Algorithms, Introduction to Machine Learning, Introduction to Computer Science, Linear Algebra & Differential Calculus, Integral Calculus, Advanced Learning Algorithms

Certifications

- **Advanced Learning Algorithms** – Stanford Online & DeepLearning.ai (Coursera)